Multi-Task Resume Analyzer HR Agent

Problem

The HR Challenge:

- HR teams read hundreds of CVs manually
- Time-consuming, prone to errors, inconsistent
- Difficult to extract structured info (skills, Q&A, improvement tips)

Our Solution:

- AI-based multi-task system that analyzes, ranks, and improves CVs automatically
- Each task handled by specialized LoRA adapter → high accuracy

Architecture Overview

RAW CVS (PDF/TXT)

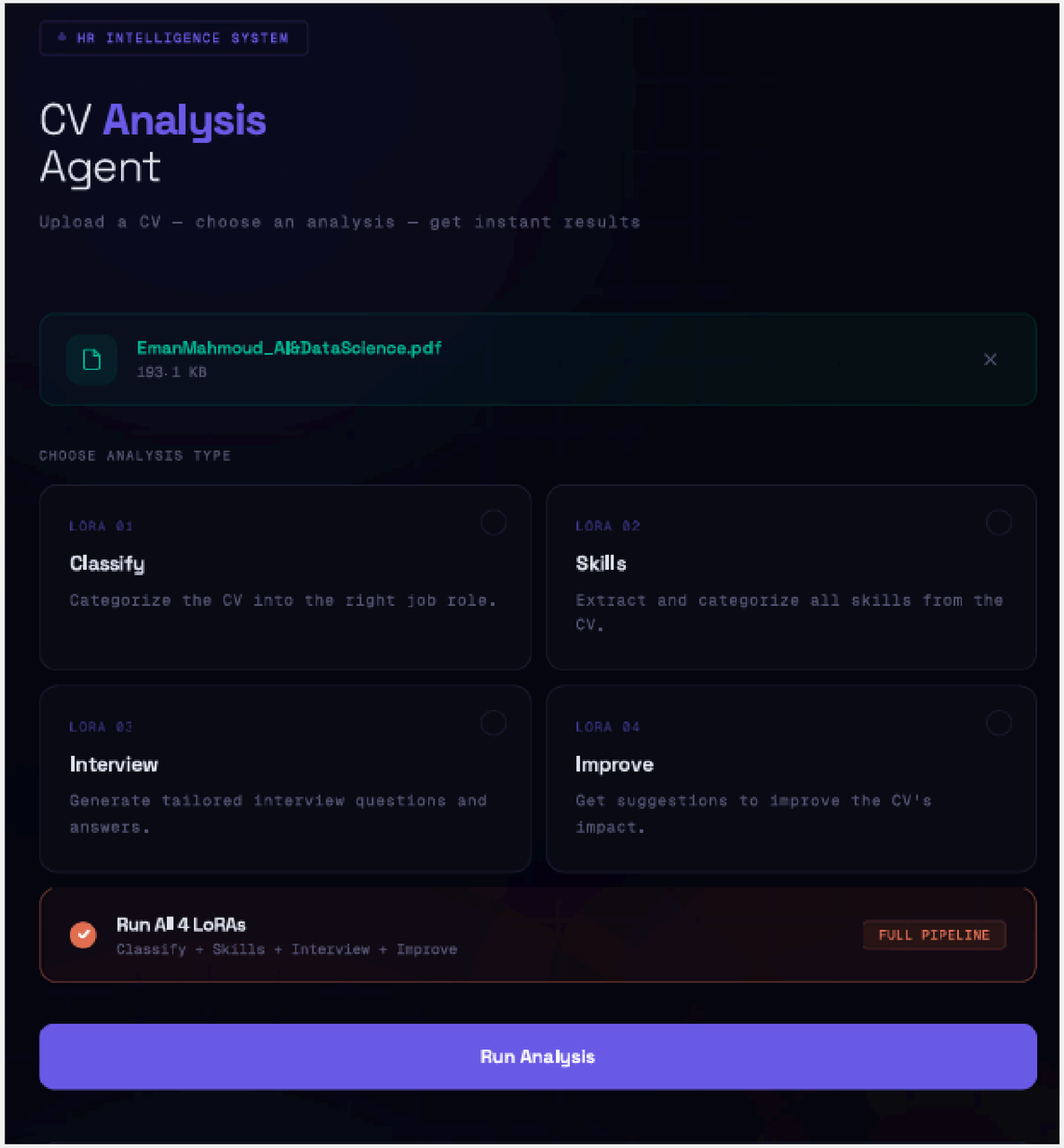
▼

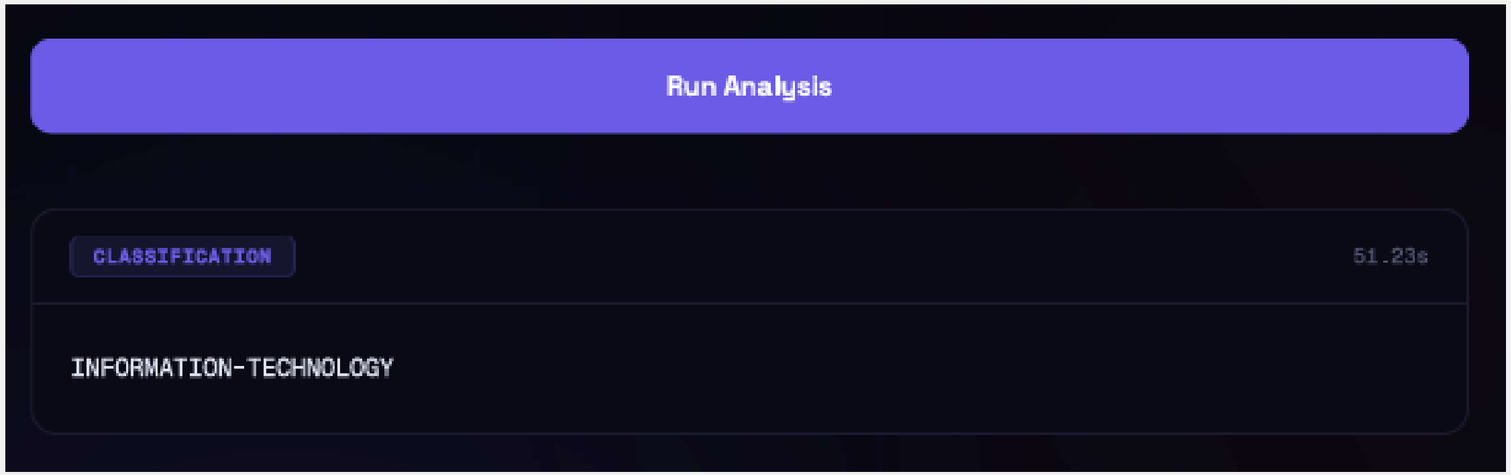
DATA EXTRACTION → LORA FINE-TUNING → DYNAMIC ADAPTER SWITCHING → API/WEB UI

- **Base model frozen (Qwen2.5-3B)**
- **4 LoRA adapters load/unload dynamically**
- **FastAPI web service for uploads, ranking, and monitoring**

How Works

- When a CV is uploaded to the HR Agent, the system first extracts all text from the PDF or TXT file using **PyMuPDF**, converting it into a clean, structured format. The base model, **Qwen2.5-3B-Instruct**, stays loaded in memory, and the system dynamically loads the required **LoRA adapter** for each task. **LoRA1** classifies the CV into the appropriate job category, **LoRA2** extracts structured skills, **LoRA3** generates suggested interview questions, and **LoRA4** provides CV improvement tips. **If multiple CVs are uploaded**, the system processes each CV with the same four adapters, calculates task-specific scores, and aggregates them to produce a f**inal ranking**, ordering the CVs from best fit to least fit. **All results**, including classification, skills, interview Q&A, and improvement recommendations, are returned in seconds through the **FastAPI** service and web UI, allowing HR teams to quickly review, compare, and make informed decisions. Integrated **Weights & Biases (W&B)** for experiment tracking, loss monitoring, and evaluation logging, and implemented runtime **monitoring** for active LoRAs, adapter switching, and CV processing statistics.


HR INTELLIGENCE SYSTEM
CV Analysis Agent
Upload a CV — choose an analysis — get instant results
EmanMahmoud_AI&DataScience.pdf
193.1 KB
CHOOSE ANALYSIS TYPE
LORA 01
Classify
Categorize the CV into the right job role.
LORA 02
Skills
Extract and categorize all skills from the CV.
LORA 03
Interview
Generate tailored interview questions and answers.
LORA 04
Improve
Get suggestions to improve the CV's impact.
Run All 4 LoRAs
Classify + Skills + Interview + Improve
FULL PIPELINE
Run Analysis


Run Analysis
CLASSIFICATION
51.23s
INFORMATION-TECHNOLOGY

SKILLS

51.23s

The candidate is a highly skilled AI Data Scientist with proficiency in programming languages like Python and SQL, and experience in various domains such as natural language processing, computer vision, and business analytics.

Technical Skills:

Python, SQL, MATLAB, TensorFlow, PyTorch, Scikit-Learn, Pandas, NumPy, Keras, C++, Hadoop, Apache Spark, OpenCV, Matplotlib, Seaborn, Jupyter Notebook, Git

Soft Skills:

Analytical thinking, Problem-solving, Team management, Communication, Time management, Leadership, Customer service

Experience:

Business Analytics Specialist, Intern: 02/2025 to 04/2025

Data Scientist Intern: 07/2024 to 08/2024

Oracle Advanced Database, Trainee Program Intern: 04/2024 to 10/2023

Educational Qualifications:

Bachelor's Degree in Data Science: Faculty of Computers and Data Science, Alexandr University, Grade: Very Good, Graduated: July 2024

INTERVIEW QS

4.36s

Q1: Can you explain a project where you applied machine learning models for predictive analytics, and how you handled data preprocessing and feature engineering?
A1: In my internship at Oasis Infobyte, I implemented supervised machine learning algorithms in Python for predictive analytics. I analyzed data to identify patterns, correlations, and anomalies. For data preprocessing, I handled missing values, performed data normalization, and feature scaling. I also applied techniques like tokenization, stopword removal, and sentiment analysis for text data. For feature engineering, I extracted relevant features from the data and used techniques like correlation analysis to select the most informative features. I then trained and evaluated machine learning models, including regression and classification models, to achieve accurate predictions.

Q2: How do you stay updated with the latest developments in AI and machine learning, and can you give an example of a recent technique you've learned and applied?
A2: I regularly follow industry blogs, research papers, and online courses to stay updated with the latest developments in AI and machine learning. For example, I recently learned about Generative AI and Prompt Engineering, and I applied these techniques to fine-tune a language model for a specific task. I also participate in online communities, like Kaggle and GitHub, to learn from others and share my own knowledge and experiences. Additionally, I've completed certifications from DataCamp and 365 DataScience to enhance my skills in machine learning and deep learning.

Q3: Can you walk us through your experience with Natural Language Processing (NLP), and how you've applied NLP techniques to real-world problems?
A3: I have experience with NLP techniques, including text preprocessing, tokenization, stopword removal, and sentiment analysis. I've applied these techniques to real-world problems, such as text classification and sentiment analysis. For example, in my internship at TechnoHacks EduTech, I conducted exploratory data analysis to identify trends within large datasets, and I assisted in developing and implementing machine learning models for predictive analytics, including NLP-based models. I've also worked with multilingual NLP, including Arabic and English, and I've used transformers like XLM-RoBERTa and BERT for text classification tasks.

Q4: How do you approach model deployment and MLOps, and can you give an example of a model you've deployed and monitored?
A4: I approach model deployment and MLOps by following a structured process, including model training, testing, and validation, as well as deployment and monitoring. I've used tools like Fast

IMPROVEMENTS

Score: 8/10

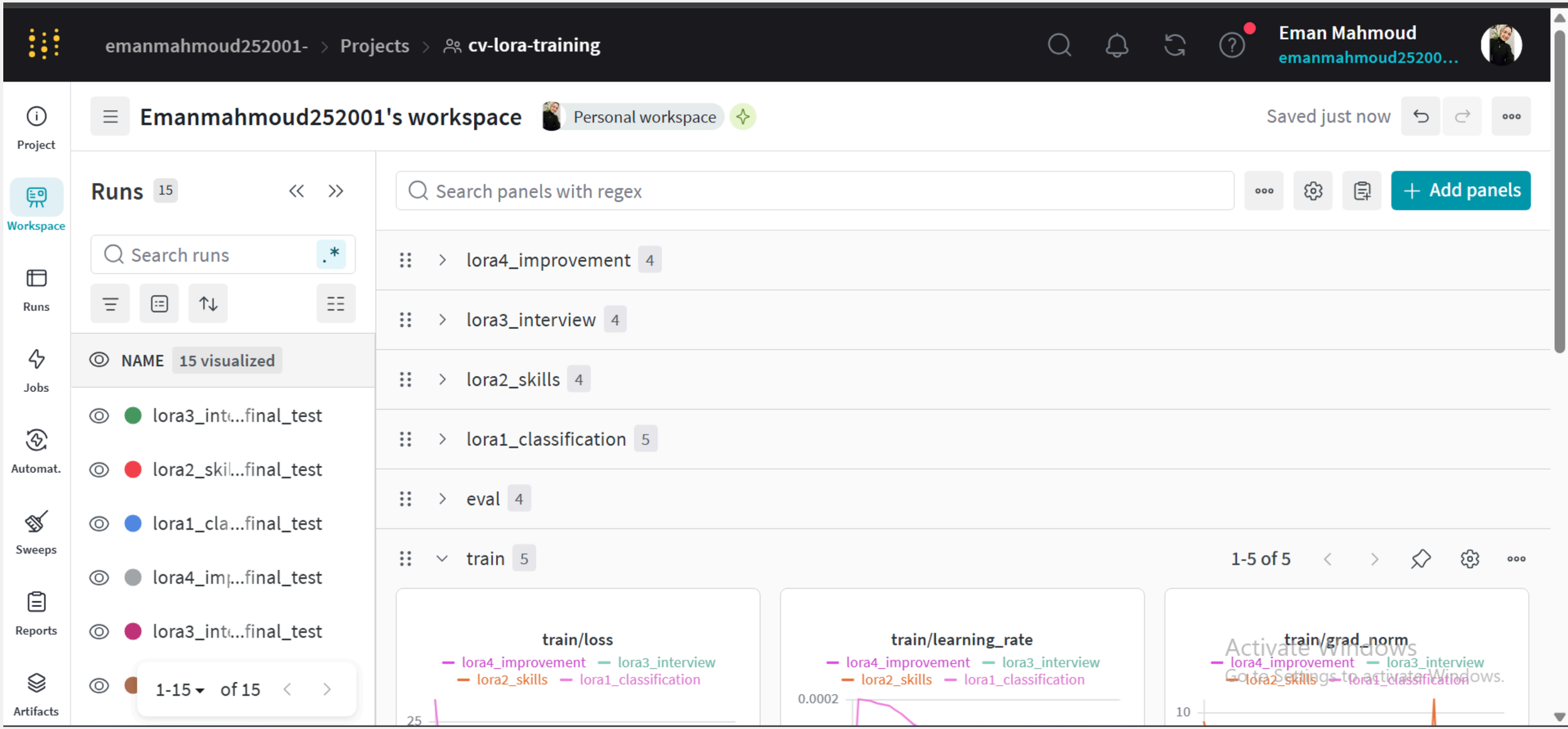
Strengths:

- Strong technical skills in Python, SQL, and natural language processing (NLP)
- Extensive experience in AI development and research
- Proven track record of delivering practical solutions through internships and projects

Improvements:

- Lack of clear career progression and job title consistency throughout the CV
- Limited use of action verbs and quantifiable achievements in work experience sections
- Incomplete education section with no GPA or other academic achievements

Tracked training, evaluation, and experiment performance using Weights & Biases (W&B) for monitoring metrics, loss curves, and model runs



wandb.ai/emanmahmoud252001-/cv-lora-training?nw=nwuseremanmahmoud252001

Project
Workspace
Runs
Jobs
Automat.
Sweeps
Reports
Artifacts

Runs 15

Search runs

NAME 15 visualized

- lora3_int...final_test
- lora2_ski...final_test
- lora1_cla...final_test
- lora4_im...final_test
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- lora2_ski...final_test
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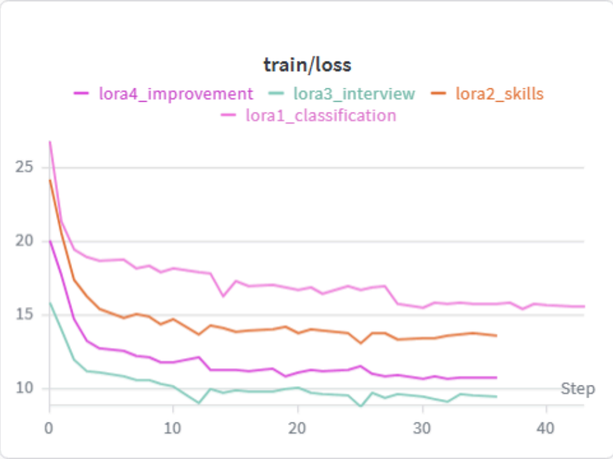
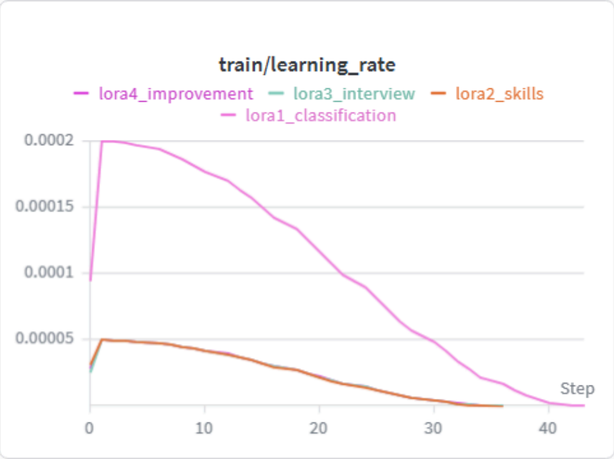
1-15 of 15

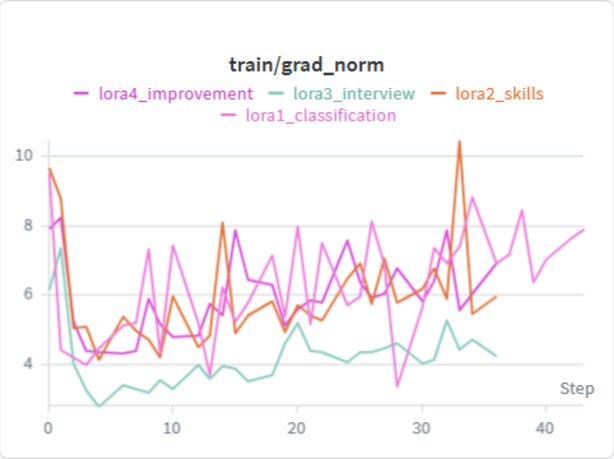
eval 4

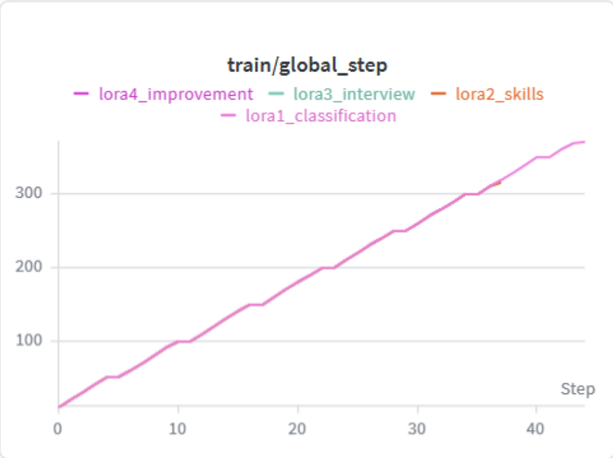
train 5

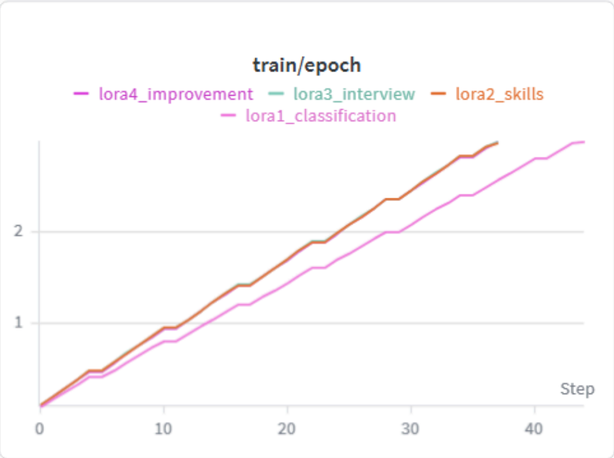
1-5 of 5











Activate Windows
Go to Settings to activate Windows.

wandb.ai/emanmahmoud252001-/cv-lora-training?nw=nwuseremanmahmoud252001

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lora1_cla...final_test

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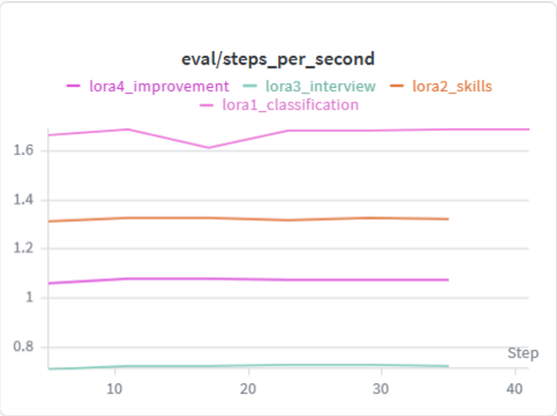
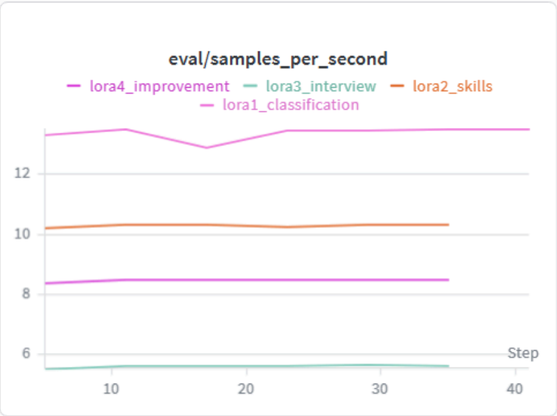
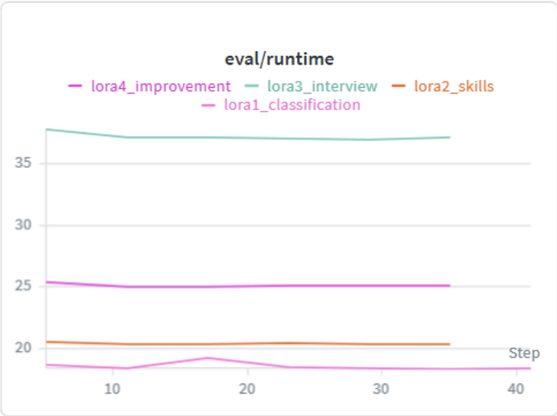
lora1_classification 5

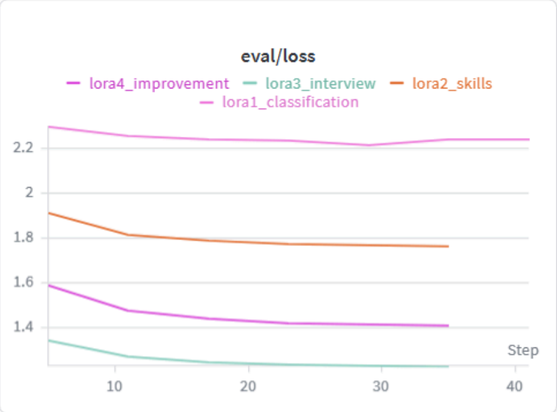
eval 4

1-4 of 4









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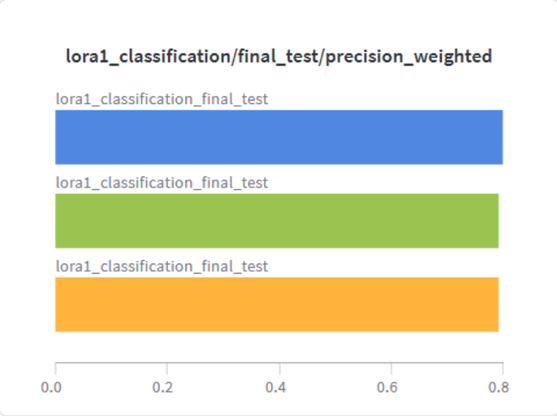
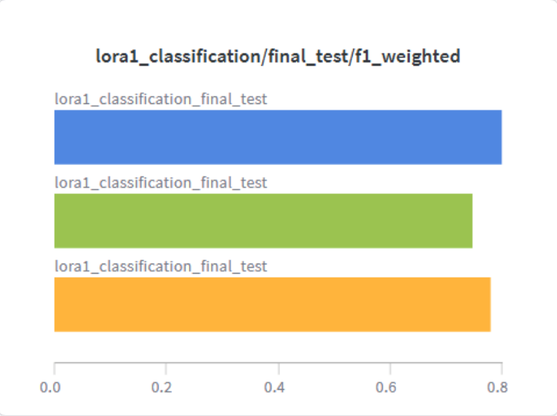
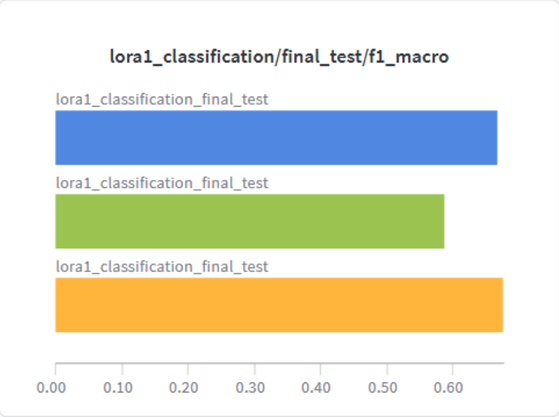
1-15 of 15

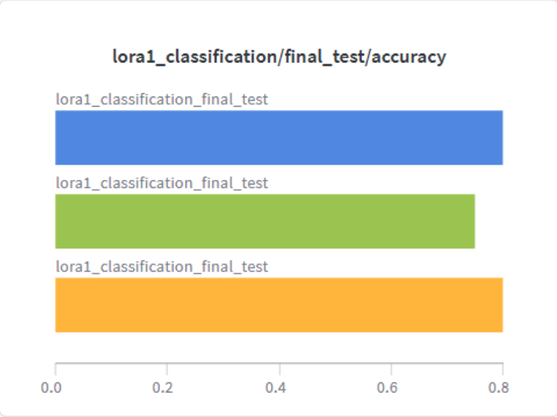
lora1_classification 5

1-5 of 5









Activate Windows
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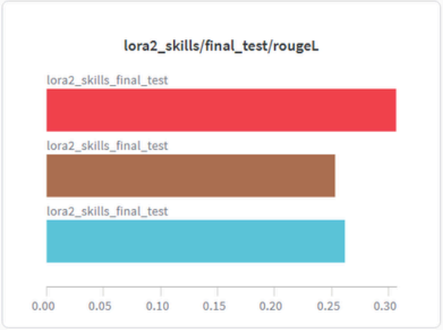
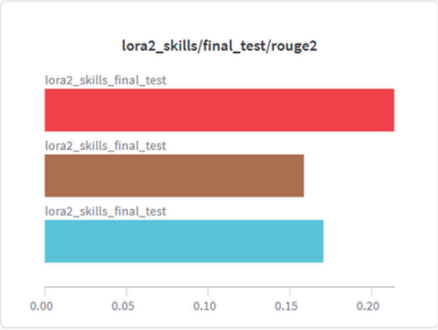
eval 4

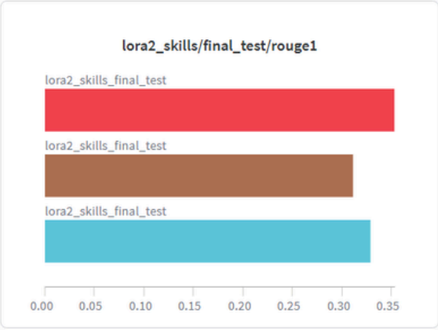
1-4 of 4

lora2_skills 4

1-4 of 4







lora3_interview 4

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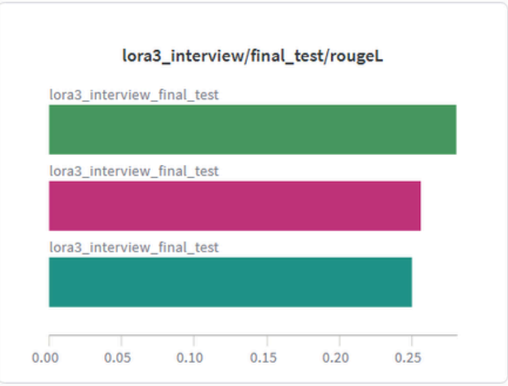

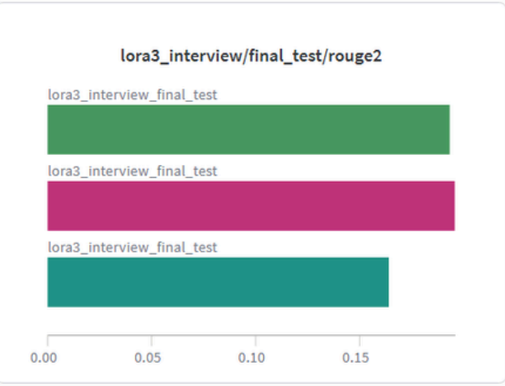

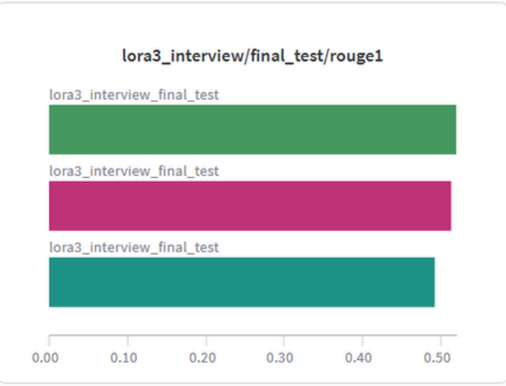

lora4_improvement 4

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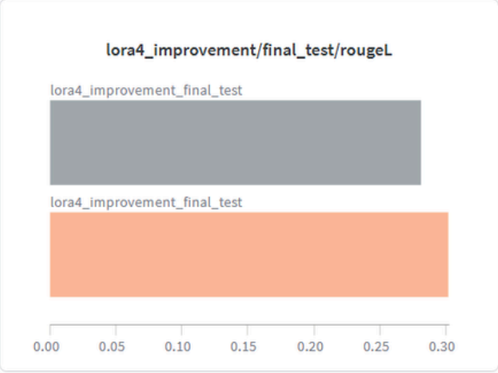
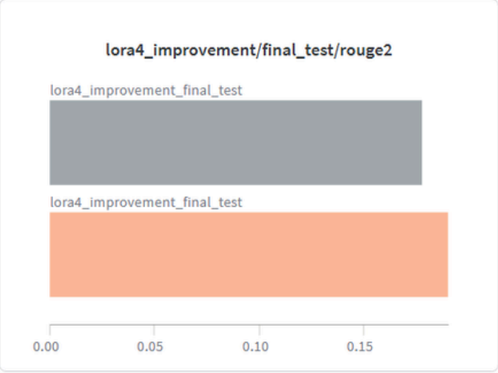
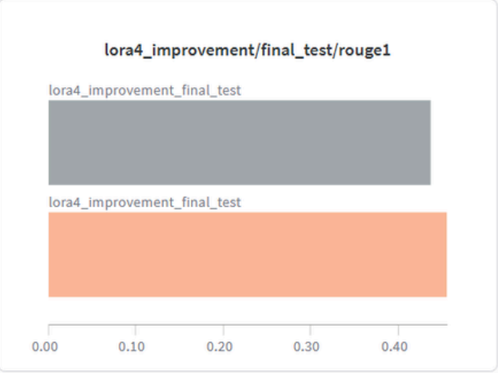




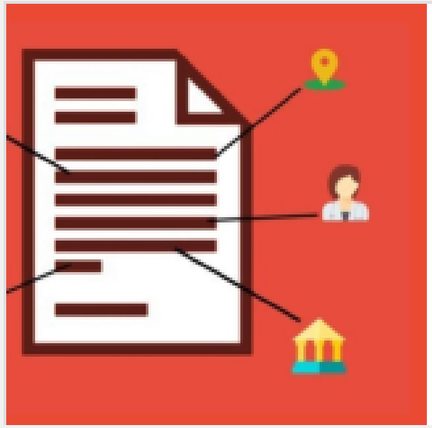


Tasks & LoRAs

Task	LoRA Adapter	Output
Classification	lora1_classification	Job category
Skills Extraction	lora2_skills	Structured skills list
Interview Q&A	lora3_interview	Suggested interview questions
CV Improvement	lora4_improvement	Resume enhancement tips

Dataset

Source: Kaggle Resume Dataset (2,400+ CVs, 25 job categories, English)
Train/Val/Test split: 80/10/10
LoRAs 2–4 training samples generated using Groq API + LLaMA-3-70B
LoRA 1 uses original job category labels



Resume Dataset

A collection of Resumes in PDF as well as String format for data extraction.

kaggle.com

Training Details

Base model: Qwen2.5-3B-Instruct (frozen)
LoRA: r=16, alpha=32, dropout=0.05
Target modules: q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj
Batch size: 2 (accumulation ×8 = effective 16)
Epochs: 3
Optimizer: AdamW, Cosine scheduler, max seq length 1024
Tracking: WandB

Github-Link

hr-agent

System Demo

System-demo